Karthik Srinivasan

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OBJECTIVE

PhD candidate in Biomedical Engineering with deep expertise in Generative AI, Computational Biology, and Statistical Physics-based ML. Passionate about applying AI to biophysical questions, including applications in healthcare. I enjoy tackling challenging problems with real-world impact and have a strong ability to quickly learn new skills. This has allowed me to contribute to diverse fields, including quantum computation, cosmology, and microbial ecology, leading to multiple publications.

EDUCATION

YALE UNIVERSITY New Haven, CT
Ph.D. in Biomedical Engineering August 2023 - Expected 2026
Focus: Analyzing microbiomes using a statistical physics based machine learning model
Relevant Coursework: Deep Learning Theory and Applications (RNNs, CNNs, LLMs, GANs, Diffusion models)

UNIVERSITY OF FLORIDA Gainesville, FL
Master of Science in Physics January 2021- August 2023
Focus: Effect of Chern-Simons coupling during inflation
CGPA: 3.91/4

INDIAN INSTITUTE OF TECHNOLOGY MADRAS Chennai, Tamil Nadu
Bachelor of Technology in Engineering Physics July 2016 - May 2020
Focus: Quantum algorithms, Schwinger effect in bouncing universes, Gravitational wave astronomy
CGPA: 8.38/10

WORK EXPERIENCE

HUMAN FRONTIER COLLECTIVE SPECIALIST - GEN AI, SCALE AI June 2025 - August 2025

- Partnered with ScaleAI researchers to rigorously evaluate **Generative AI (LLM)** performance.
- Designed complex, domain-specific challenge sets to identify model hallucinations and reasoning failures, directly informing model fine-tuning strategies.

RESEARCH EXPERIENCE

GRADUATE RESEARCH ASSISTANT, YALE UNIVERSITY August 2023 - Present

Identifying environmental dimensionality of microbiomes using Machine Learning

- Analyzed **NGS 16S rRNA** datasets from Microbiomap using **machine learning**
- **Devised and validated a metric to quantify dimensionality** of microbiomes based on model performance
- Implemented **Markov Chain Monte Carlo (MCMC)** methods for high-dimensional parameter space exploration, optimizing model performance on large-scale biological datasets.
- **Built an open source reproducible codebase** for simulation and analysis microbiomes.
- DOI: https://doi.org/10.1101/2025.11.02.686095 (under review)

Designing host-associated microbiomes using the consumer-resource model

- Developed a **generative mechanistic machine learning model** to analyze host-microbe health studies.
- Performed **sample specific statistical analysis** via bootstrapping of generative model
- Biased sampling of the latent space via **Monte Carlo methods** to explore unobserved phenotypic regions
- Developed a production-ready codebase **currently adopted by industry (Biomedit) for therapeutic discovery**.
- DOI: https://doi.org/10.1128/msystems.01068-24, No. of citations: **7**

GRADUATE RESEARCH ASSISTANT, UNIVERSITY OF FLORIDA January 2021 - July 2023

Gravitational wave probes of massive gauge bosons at the cosmological collider

- Leveraged **High-Performance Computing (HPC)** clusters to perform large-scale numerical integrations for cosmological simulations.
- Using the in-in formalism in Quantum Field Theory in Curved Spacetime, we computed the two-point and three-point correlation functions for both scalar and tensor perturbations
- Developed parallelized simulation pipelines to model complex theoretical physics phenomena
- DOI: https://doi.org/10.1088/1475-7516/2023/02/013, No. of citations: **40**

Parity-odd and even trispectrum from axion inflation

- Extended the in-in formalism to compute the four-point function to study parity-violation in early universe
- Chern-Simons interaction could explain baryon asymmetry which is captured in the parity odd four-point function
- Using Numerical Integration with High-performance Cluster-computers we found parity-odd signal is non-vanishing
- DOI: https://doi.org/10.1088/1475-7516/2023/05/018, No. of citations: **59**

UNDERGRADUATE RESEARCH INTERN, IISER KOLKATA Summer/Winter 2018

Efficient quantum algorithm for solving traveling salesman problem: An IBM quantum experience

- **Developed a quantum algorithm** to solve the traveling salesman problem on a quantum computer
- **Implemented the algorithm on a quantum computer** via IBM quantum experience (now IBM quantum platform)
- The algorithm theoretically achieves a **quadratic speedup in compute time** over the best classical algorithms
- DOI: https://doi.org/10.48550/arXiv.1805.10928, No. of citations: **149**

Nondestructive discrimination of a new family of highly entangled quantum states

- Proposed a new set of highly entangled states which can be used as an orthonormal bases of error correction and measurement based quantum computing
- **Designed circuits** using IBM quantum experience to produce and distinguish these states non-destructively
- Using Quantum State Tomography we showed that the states were preserved
- DOI: https://doi.org/10.1007/s11128-018-1976-9, No. of citations: **67**

PERSONAL PROJECTS

Generative AI for Drug Design: Protease-Resistant Insulin | RFdiffusion, Pepsickle Spring 2025

- Mapped protease-cleavage hotspots on human insulin using **Pepsickle** and structural **PDB** references.
- Utilized **RFdiffusion (Generative AI)** to engineer carrier proteins for insulin delivery, optimized for protease resistance.
- Designed and validated 3D protein structures using **computational biology pipelines**, demonstrating the application of GenAI in therapeutic discovery: https://github.com/karthik-yale/ProteaseResistantInsulin

Thermodynamic Manifold Inference Fall 2022

- **Re-implemented** the Thermodynamic Manifold Inference (TMI) algorithm from the original Phys. Rev. Research paper, including gradient-based training, out-of-sample embedding, and performance logging.
- **Applied** the model to the full MNIST dataset (70 k images), compressing 784-dimensional inputs to a 2-D manifold while retaining high reconstruction fidelity.
- **Released** a open-source codebase and tutorial notebook: https://github.com/karthikvasan99/TMI_MNIST

Higgs Boson Classification Challenge | Scikit-learn Fall 2022

- **Engineered** feature pipelines (polynomial basis expansion, PCA) for CERN's Higgs Boson Machine-Learning dataset.
- **Benchmarked** Support Vector, Random Forest, and Gradient Boosting classifiers; tuned hyper-parameters via exhaustive grid search to maximize ROC-AUC.
- **Published** reproducible code and results: https://github.com/karthikvasan99/higgs-boson-classification

ADDITIONAL

Languages: Python, C++, SQL, R. **ML:** PyTorch, TensorFlow, Scikit-learn. **GenAI & LLM tools:** CrewAI, RFdiffusion.
Scientific Computing: Numpy, Pandas, SciPy, Matplotlib, HPC (Slurm)